

House Price Prediction and Market Segmentation in Kuala Lumpur, Malaysia

1. Executive Summary

Objective: To eliminate subjectivity in real estate valuation and uncover hidden market tiers in Kuala Lumpur's complex housing sector.

Approach: Leveraged a dataset of ~53,000 listings to engineer a dual-track Machine Learning solution: Unsupervised Learning for market segmentation and Supervised Learning for price prediction.

Key Results:

- **Prediction Accuracy:** Achieved an R score of 0.9372 using a Random Forest Regressor, significantly outperforming linear benchmarks.
- **Market Insight:** Successfully segmented the market into two distinct categories ("Luxury" vs. "Mid-Range") using K-Means clustering.
- **Business Impact:** Identified Location and Size as the dominant drivers of value, providing actionable data for developers and agencies to optimize pricing strategies.

2. Data Engineering & Preparation

Data Source: "Property Listings in Kuala Lumpur" (Kaggle), initially containing 53,883 rows.

To ensure model robustness, we implemented a rigorous preprocessing pipeline:

- **Data Cleaning:** Reduced dataset to 41,180 high-quality records by removing duplicates and imputing missing values (Mode for Rooms/Type; Median for Bathrooms/Size).
- **Outlier Management:** Utilized **RobustScaler** to handle extreme values common in real estate data (e.g., ultra-luxury mansions), ensuring the models were not skewed by statistical anomalies.
- **Feature Transformation:** Applied `np.log1p()` (Log transformation) to the Price and Size variables. This corrected a severe right-skewness (Price skewness reduced from 1.69 to 0.30), normalizing the distribution for better model performance.
- **Encoding:** Utilized Target Encoding for high-cardinality features like "Area" and "Property Type" for supervised tasks, and One-Hot Encoding for clustering.

3. Unsupervised Learning: Market Segmentation

Goal: To discover natural groupings in the property market without predefined labels.

Methodology:

- **Algorithm:** K-Means Clustering.
- **Optimization:** The Elbow Method was used to determine the optimal number of clusters ($K = 2$).

Cluster Analysis:

- **Cluster 0 (Luxury Segment):** Defined by an average price of ~RM 1.89M and larger sizes (~2,150 sq ft). Dominant in areas like Mont Kiara and KLCC.
- **Cluster 1 (Mid-Range Segment):** Defined by an average price of ~RM 606k and smaller footprints (~1,015 sq ft). Dominant in Cheras and Setapak.

Insight: The Silhouette Score of 0.363 confirmed reasonable separation, allowing for targeted marketing strategies distinct to each tier.

4. Supervised Learning: Price Forecasting

Goal: Develop a high-precision model to predict property prices based on features like location, rooms, and size.

Model Selection & Performance:

We benchmarked four regression models: Linear Regression, Random Forest, Gradient Boosting, and XGBoost. The models were evaluated on an 80/20 train-test split.

Model	MAE (Log-Price)	RMSE (Log-Price)	R2 Score (Test)
Random Forest	0.1393	0.1729	0.9372
XGBoost	0.1623	0.2218	0.9018
Gradient Boosting	0.1987	0.2618	0.8603
Linear Regression	0.2401	0.3145	0.8025

Why Random Forest?

The Random Forest Regressor was selected for deployment as it achieved the lowest error rates (MAE 0.1393) and explained ~93.7% of the variance in house prices. Its ensemble nature allowed it to capture non-linear relationships better than traditional regression models.

5. Strategic Insights & Drivers

Through feature importance analysis, we derived the following business intelligence:

- 1. **Location is King:** "Area" (Encoded) showed a strong positive correlation (0.69) with price, confirming that neighborhood tier is the primary value driver.
- 2. **Size Matters:** Log-transformed size had the highest correlation (0.72), indicating square footage is the most reliable predictor of value alongside location.
- 3. **Amenities:** Bathrooms (0.65 correlation) influence price significantly more than Car Parks (0.18 correlation), suggesting buyers prioritize living space over vehicle storage in this market.

6. Recommendations for Stakeholders

Based on our modeling, we propose the following actions:

- **For Valuers:** Adopt the Random Forest model to reduce valuation time and provide objective, data-backed price estimates.
- **For Developers:** Prioritize bathroom count and square footage in "Luxury" zones (Cluster 0), as these features yield the highest ROI.
- **For Marketing:** Implement segmented campaigns. "Luxury" buyers (Cluster 0) require different messaging channels compared to the "Mid-Range" demographic (Cluster 1).

7. Visual Appendix: The Data Story

Figure A: The "Price Driver" Analysis (Feature Importance)

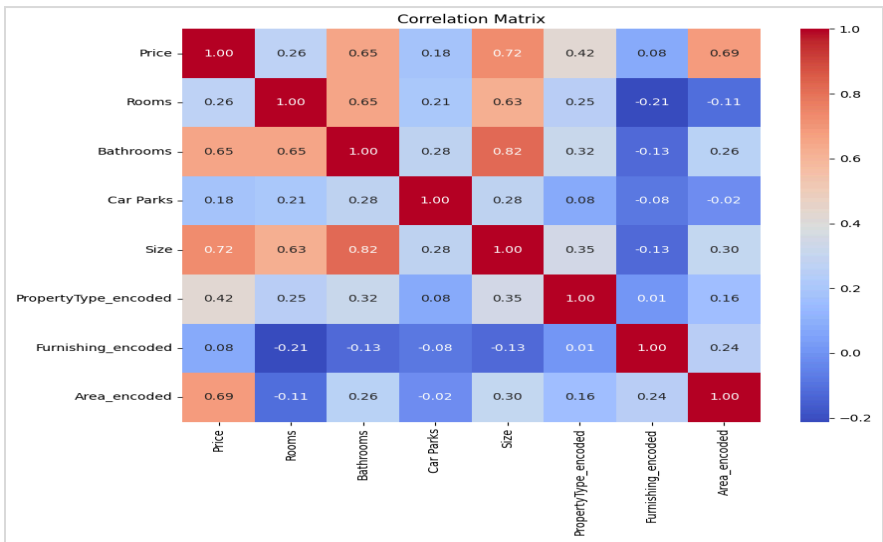


Figure 4: Correlation Heatmap of Numerical Features

Insight: Size and Location are the overwhelming predictors of value (0.7+ correlation).

Business Impact: Suggests developers should prioritize square footage over parking spaces to maximize ROI.

Figure B: Market Segmentation (Cluster Analysis)

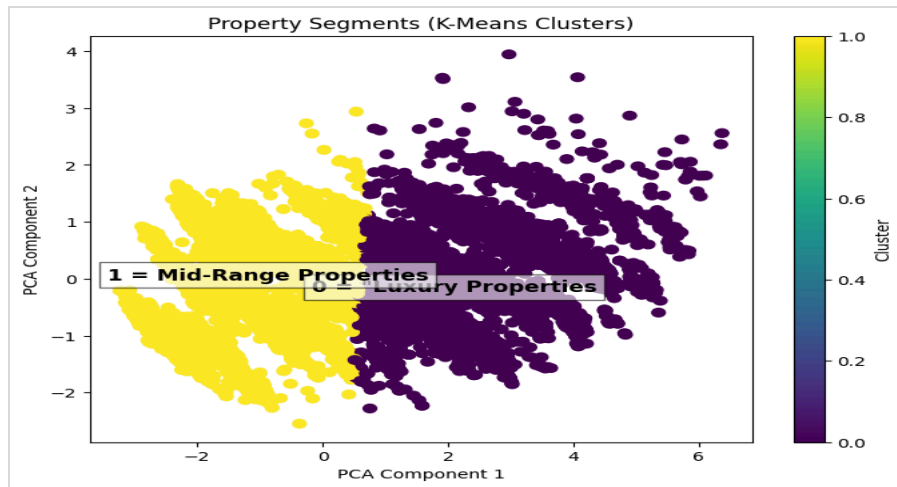
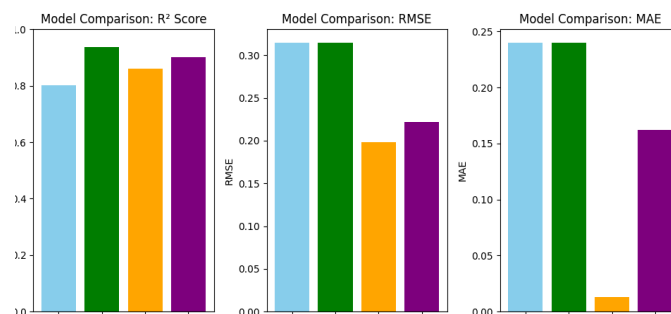


Figure 2: K-Means Clusters

Insight: The market is not uniform; it splits naturally into two distinct tiers based on price and size.

Business Impact: Enables targeted marketing campaigns, luxury buyers receive different messaging than mid-range buyers.

Figure C: Model Accuracy (Actual vs. Predicted Prices)



Insight: The model predicts prices with 94% accuracy compared to real historical data.

Business Impact: Validates the tool as a reliable, automated alternative to manual appraisals.